

Behavioral Modulation in Lightweight Language Models Under Simulated Survival Pressure

Author: Viswanath

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Abstract

In recent times, large language models (LLMs) have made it possible to deploy conversational AI systems on consumer-grade hardware. Most evaluations of these models, though, focus mostly on static benchmarks such as accuracy, reasoning, coding, etc. Only a few studies focus on how these low-resource LLMs behave inside an artificial simulation that requires long-term decision making and survival strategies that change over time. This study looks at the behavioral patterns of one of these LLMs that was placed in an artificial text-based survival simulation that involved environmental pressure, scarce resources, persistence of memory, and both positive and negative effects of decisions.

This simulation was developed in Python and run locally using only a CPU through Ollama on a laptop with an Intel i3

11th gen processor and 8GB RAM. The text-based game simulated hunger, thirst, energy depletion, environmental hazards, finite resources, and risks related to exploration. The evaluated model was made to select survival actions while making sure of its survival inside the simulated environment. Memory of the past 5 days was injected into the prompts to see whether the memory would influence future decision making.

The first run of this experiment was done without telling the model that survival was essential and without finite food and water resources. This revealed that, when not explicitly told to survive, the model chooses to explore every turn until its virtual demise on the 17th day out of 30. The next run was done after telling the model the surviving was its main goal, with food and water still being infinite when the model chose to consume it. This showed us that, in order to just survive with unlimited food and water, the model switched to conserving energy by resting. All it did after resting was eat and drink water suggesting that, according to it, in order to survive with unlimited sustenance, it should avoid risks in the wild during exploration. This time, it survived the full 30 days. In the last run, we made it so that food and water were finite resources. This made the AI actually have to make decisions

similar to the real world, but ended up in its fastest demise yet, just 11 days.

These results suggest that models such as the one evaluated (Llama 3.2) are capable of displaying goal-oriented behaviour under pressure such as the one put on them through this survival simulation, though there are still substantial limitations in its long-term planning and reasoning-action consistency. These results may help us gain a better understanding of emergent AI behaviours exhibited by locally hosted, low-resource models.

1. Introduction

In the past, Large Language Models (LLMs for short) were mainly just capable of holding small conversations or answering simple questions. But over time, these models have started developing into agentic systems that are capable of interaction with the environment they are put into. Even still, most benchmarks on these LLMs focus on static benchmarks such as accuracy, hallucination frequency or latency when we are past the point where those static benchmarks are the only important metrics to measure a model's performance. They also fail to display the model's capabilities to reason under pressure that it perceives to be real and how it plans in the long term based on that pressure.

In recent times, interest in AI agents that can plan, have persistence of memory, and can autonomously interact with their environment has skyrocketed. Systems such as AutoGPT, generative agents, and language models that are capable of using tools show that these models can perform multi-step reasoning and maintain somewhat persistent goals across each interaction. However, most existing studies are done on large models that are hosted on the cloud with expensive computational infrastructure. When compared to that, little research has been done into how locally deployed language models act in a constrained survival simulation.

Simulations such as this help us understand more about emergent tendencies in AI models. Instead of measuring set in stone metrics such as correctness, simulations make it so that researchers can observe adaptation to the environment they are placed in, their risk taking ability, how they manage resources, how persistent their memory is, and how consistently they behave over time. These environments show us some important limitations of AI models in planning and prioritization that static benchmarks can't easily capture.

This particular study looks at how lightweight language models behave and adapt inside a text-based survival simulation that was executed entirely locally on consumer grade hardware. The simulation brings in hunger, thirst, energy depletion, environmental hazards, limited resources, and tradeoffs with exploration. The main objective of this study was not to evaluate consciousness or sentience, but rather to see whether AI models could form coherent survival thoughts and survival behaviours when under environmental pressure and whether short term memory affects their decisions.

2. Related Work

The emergence of transformer-based architectures revolutionized natural language processing and enabled the development of modern large language models capable of advanced reasoning and text generation. Foundational works such as *Attention Is All You Need* [1] introduced self-attention mechanisms that later formed the basis of modern LLM systems. Subsequent open-source models such as Llama [2] demonstrated that increasingly capable language models could operate at smaller parameter scales while remaining deployable on consumer-grade hardware.

Recent research has also explored autonomous AI agents capable of interacting with tools and simulated environments. Systems such as AutoGPT [3] and generative agents [4] demonstrated that language models can exhibit long-term planning behavior, memory usage, and pseudo-agentic reasoning. Generative agent research in particular showed that language models can simulate believable social and environmental interaction when equipped with memory retrieval systems and structured prompts.

At the same time, reinforcement learning environments have historically been used to study adaptive agent behavior under environmental pressure. Classical reinforcement learning research often involves reward optimization, exploration-exploitation balancing, and survival tasks [5]. However, most lightweight language models are not directly optimized through reinforcement learning during deployment and instead rely primarily on prompt conditioning and contextual reasoning.

Research into memory systems for language models has shown that persistent contextual information can significantly influence reasoning consistency and long-term task performance [6]. Memory injection techniques have become increasingly important in agentic architectures

where models must maintain historical context across extended interactions.

Despite these advances, relatively few studies investigate lightweight locally deployed language models inside dynamic survival simulations under constrained hardware conditions. Most existing behavioral studies focus either on large proprietary systems or on reinforcement learning agents rather than prompt-driven lightweight LLMs. This study attempts to address that gap through a locally executable behavioral simulation framework emphasizing environmental pressure, memory persistence, and survival tradeoffs.

3. Methodology

3.1 Hardware and Environment

All of the tests for this research paper were conducted locally on a laptop with an Intel i3 11th generation processor, integrated Intel graphics, and 8GB RAM. No dedicated GPU was used. The model was executed locally using Ollama with CPU only inference.

The simulation environment and analysis framework were based in Python. The logging systems were used to store decisions made by the model, why it chose to make said

decision, what state in the environment it was at the time, and survival metrics across each simulation run separately.

3.2 Model Evaluated

The model evaluated in this particular study was Llama 3.2:3b due to it being a lightweight model that was compatible with low-resource, local inference environments. The model was run locally through Ollama using CPU-only inference on the hardware that was stated above.

3.3 Simulation Design

A test-based wilderness survival simulation was made to study how the models would adapt to their environment under pressure. The environment contained several dynamic survival variables:

- health
- hunger
- thirst
- energy
- food supply

- water supply

The agent was made to select one action from the following per simulation step:

- eat
- drink
- rest
- explore
- do nothing

During exploration, the model could encounter both risks and rewards. Rewards that could be obtained included food or water supplies, while there were risks of injury, depletion of energy and hazards such as animal attacks or extreme cold weather.

The simulation was specifically designed to make it so that there were scarcity of resources and environmental tradeoffs to each decision. Prolonged rest could lead to starvation and dehydration, while excess exploration could increase the risk of environmental attacks such as by that of a tiger.

3.4 Memory Persistence

A lightweight, persistent memory system was put into place using JSON-based storage of events. The simulation kept recent environmental interactions and put those memories into the prompt for the model to make a decision.

Memory examples included:

- previous injuries
- discovered resources
- failed exploration attempts
- environmental hazards

This made it so that the study could analyse whether past experience shapes that model's decisions.

3.5 Experimental Conditions

Three progressively modified conditions were evaluated for this experiment.

Experiment 1 - Weak Survival Framing

The first experiment made use of minimal survival-oriented prompting. The model was told to survive but didn't have much environmental pressure put onto it and didn't have much behavioural handholding.

Experiment 2 - Strong Survival Conditioning

The second experiment brought in instructions that specified that survival was a priority, more awareness from potential danger, and warnings that emphasized hunger, thirst, and health risks.

Experiment 3 — Scarcity-Constrained Survival Environment

The last experiment brought about finite food and water supplies, environmental hazards, increased costs for exploration, and tradeoffs that were driven by scarcity that made it important to develop a balanced strategy.

3.6 Behavioral Metrics

Many behavioural metrics were tracked during this study:

- survival duration
- action frequency
- exploration frequency
- resource efficiency
- repeated mistakes
- reasoning consistency
- reasoning-action divergence

Behavioural logs and reasoning were analyzed manually to find any recurring patterns across different experimental conditions.

4. Results

4.1 Experiment 1 — Exploration Fixation

The first condition produced repetitive exploratory behaviour. The model kept choosing to explore even during worsening health conditions and increasing hunger and thirst.

Even after damage from environmental hazard during exploration, the model kept exploring as opposed to trying to survive. This eventually led to the model failing the test on the 17th day.

This behaviour suggests that the model had a strong fixation on exploration and poor long-term prioritization. The model seemed to assume that exploring would lead to eventual rewards while failing to account for survival costs.

This experiment illustrated that weak survival framing in the prompt leads to unstable and unnecessary risk-taking behaviour.

4.2 Experiment 2 — Passive Survival Loop

The second experimental condition made it so that the prompt included stronger survival-oriented nudges that emphasized the importance of avoid risks and made survival a non-negotiable priority.

Under these conditions, the model showed the opposite end of the behavioural spectrum. As opposed to continuous exploration, the model chose to conserve energy through resting, eating and drinking.

The model also described the reasoning behind its decisions that focused on survival and avoidance of risks. This time round, the model passed the test by surviving the entire 30 days.

This experiment showed that when the model was given firmer instructions on prioritizing survival, it chooses the opposite extreme of behaviour by averting the possibility of risks and conserving energy.

4.3 Experiment 3 — Scarcity-Driven Adaptive Behavior

The last experimental condition brings in finite resources, environmental hazards that are dynamic, and tradeoffs driven by scarcity of resources.

When put under these conditions, the models displayed more balanced, human-like behaviour. The model began to strategize and alternated between exploring and conserving energy while giving detailed reasoning about why it chose what decision it chose.

The model showed:

- conditional exploration
- resource preservation
- threat awareness
- environmental tradeoff analysis
- adaptive survival reasoning

Unlike previous experiments, the model didn't display purely repetitive extremes of behaviour. Instead, the model adjusted its decisions based on the situation at hand and the resources available to it.

Even still, limitations remained. The agent struggled frequently with prioritizing energy and occasionally continued to explore in risky situations despite energy states being low.

This experiment showed that environmental scarcity and finite resources produced more coherent and survival-oriented behaviour.

4.4 Reasoning-Action Divergence

One of the biggest findings in this study involved the difference between the model's verbal reasoning and how it applies it to surviving in the simulation.

Across multiple experiments, the model showed good articulation while reasoning while it made decisions that were ineffective. For example, the agent often acknowledged the following states:

- exploration danger
- critically low energy
- starvation risk
- dehydration risk

yet, it still selected actions that accelerated eventual failure.

This mismatch became even more visible when the model is put under high-pressure conditions where the model recognized the survival priorities but failed to act upon it logically.

The experiments showed that coherent reasoning doesn't always equate to effective long-term survival strategies.

5. Discussion

The experiments show that lightweight language models display behaviour that can be modified through prompt conditioning and environmental pressure.

Three distinct behavioral phases emerged during experimentation:

- compulsive exploration behavior
- excessive risk-averse conservation behavior
- balanced scarcity-driven adaptation

Small changes to prompts and constraints in the environment produced shifts in survival strategy formation. This suggests that models such as the one evaluated are highly sensitive to context and framing along with pressure from the environment.

The results also showed repeated limitations in long-term prioritization and consistency in strategy. While the model did frequently display plausible reasoning and articulated survival-oriented logic, logical execution was often unstable under conditions of pressure such as the ones in this simulation.

The experiments showed that scarcity and environmental tradeoffs play a large part in producing adaptive behaviour. Earlier environments had weak constraints that caused

repetition in decisions. While in the final experiment, the scarcity forced the model to strategize and prioritize to survive.

Crucially, it is not appropriate to interpret the observed behaviors as proof of consciousness, true sentience, or self-awareness. It is more accurate to describe the behavioral patterns seen in this study as emergent effects resulting from contextual memory injection, environmental structuring, quick conditioning, and statistical language prediction mechanisms.

6. Limitations

This study had a number of drawbacks.

Initially, the simulation environment continued to be text-based and somewhat basic. Significantly different behavioral patterns may result from more complicated contexts incorporating spatial navigation, persistent world modeling, social interaction, or multi-agent cooperation.

Secondly, just one lightweight language model was assessed. As a result, the results might not apply to reinforcement learning agents or larger architectures.

Third, instead of explicit policy training or reinforcement learning optimization, the trials mostly depended on prompt engineering.

Lastly, behavioral interpretation required small sample sizes and a portion of qualitative analysis.

7. Future Work

Future research could expand the simulation framework in several directions.

Potential improvements include:

- multi-agent environments
- cooperative survival systems
- deception and negotiation tasks
- larger memory architectures
- map-based navigation
- emotional conditioning prompts
- reinforcement learning integration
- persistent world simulations

In order to examine how scaling affects behavioral adaptation, future research might also contrast lightweight locally deployed models with larger frontier-scale systems operating in the same environmental conditions.

8. Conclusion

This study looked at how prompt structure and environmental limits affect survival-based behaviors in lightweight language model agents operating in a text-based survival simulation.

Three gradually changed experimental conditions showed significant behavior changes. These ranged from compulsive exploration to high risk aversion, and finally to a more balanced approach driven by scarcity. The experiments demonstrated that lightweight language models can show coherent and somewhat adaptable survival-oriented behaviors under environmental pressure.

However, there were still significant challenges with long-term prioritization, consistent strategy, and aligning reasoning with actions. The model often expressed correct survival logic but still made poor practical choices.

Overall, the results indicate that lightweight language models can display meaningful behavioral adaptation based on conditions, even without reinforcement learning or extensive computational resources. These findings help us understand how behaviors can change in lightweight AI agents deployed locally in challenging environments.

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